Batch: A2 Roll No.: 16010324044
Experiment / assignment / tutorial No. 10
Grade: AA / AB / BB / BC / CC / CD /DD
Signature of the Staff In-charge with date

TITLE: Configure DNS server using Cisco Packet Tracer.

AIM: To configure DNS server for given network topology using Cisco Packet Tracer.

OUTCOME: Student will be able to Understand operations of various application layer protocols.

Theory:

The Domain Name System (DNS) is a hierarchical and decentralized naming system for computers, services, or other resources connected to the Internet or a private network. It associates various information with domain names assigned to each of the participating entities. Most prominently, it translates more readily memorized domain names to the numerical IP addresses needed for locating and identifying computer services and devices with the underlying network protocols. By providing a worldwide, distributed directory service, the Domain Name System has been an essential component of the functionality of the Internet.

The Domain Name System delegates the responsibility of assigning domain names and mapping those names to Internet resources by designating authoritative name servers for each domain. Network administrators may delegate authority over sub-domains of their allocated name space to other name servers. This mechanism provides distributed and fault-tolerant service and was designed to avoid a single large central database.

The Domain Name System also specifies the technical functionality of the database service that is at its core. It defines the DNS protocol, a detailed specification of the data structures and data communication exchanges used in the DNS, as part of the Internet Protocol Suite.

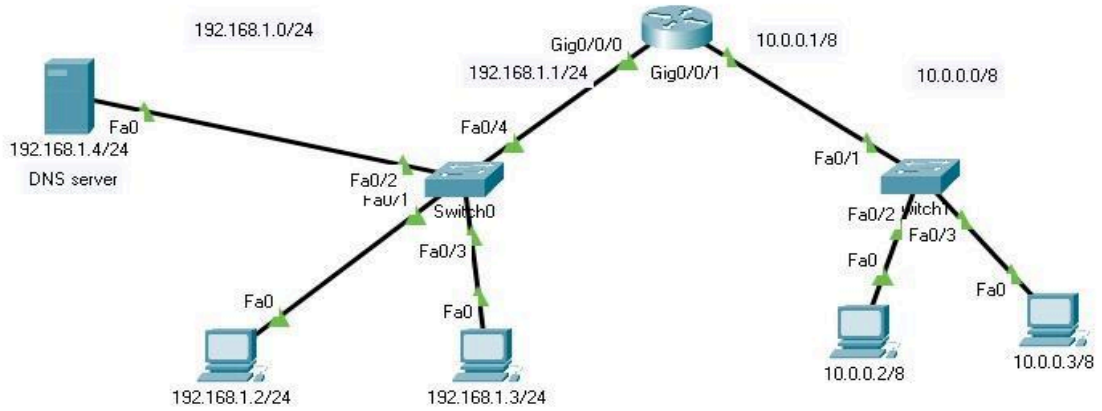
The Internet maintains two principal namespaces, the domain name hierarchy and the Internet Protocol (IP) address spaces. The Domain Name System maintains the domain name hierarchy and provides translation services between it and the address spaces. Internet name servers and a communication protocol implement the Domain Name System. A DNS name server is a server that stores the DNS records for a domain; a DNS name server responds with answers to queries against its database.

The most common types of records stored in the DNS database are for Start of Authority (SOA), IP addresses (A and AAAA), SMTP mail exchangers (MX), name servers (NS), pointers for reverse DNS lookups (PTR), and domain name aliases (CNAME). Although not intended to be a general purpose database, DNS has been expanded over time to store records for other types of data for either automatic lookups, such as DNSSEC records, or for human queries such as responsible person (RP) records. As a general purpose database, the

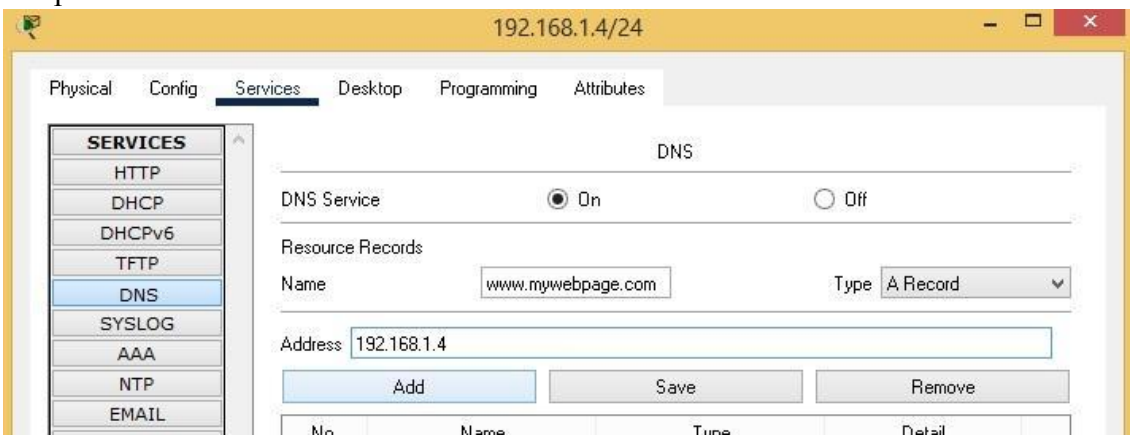
DNS has also been used in combating unsolicited email (spam) by storing a real-time blackhole list (RBL). The DNS database is traditionally stored in a structured text file, the zone file, but other database systems are common.

Procedure:

1. To begin with the experiment, open Cisco Packet Tracer
2. Create the lab and configure the basic IP addressing for all PCs and server for the network topology shown below:



3. Open the services tab in server and make DNS service ON



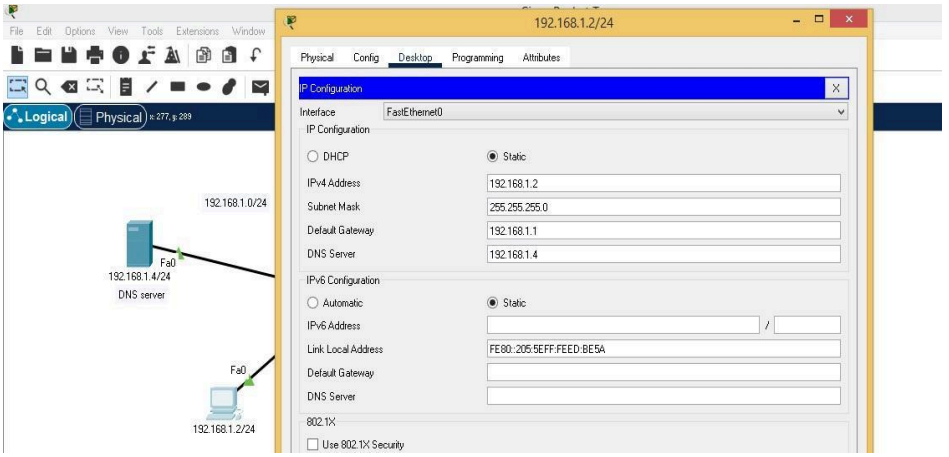
Add name as www.mywebpage.com and put IP address as 192.168.1.4

Then click on ADD button.

Close the window

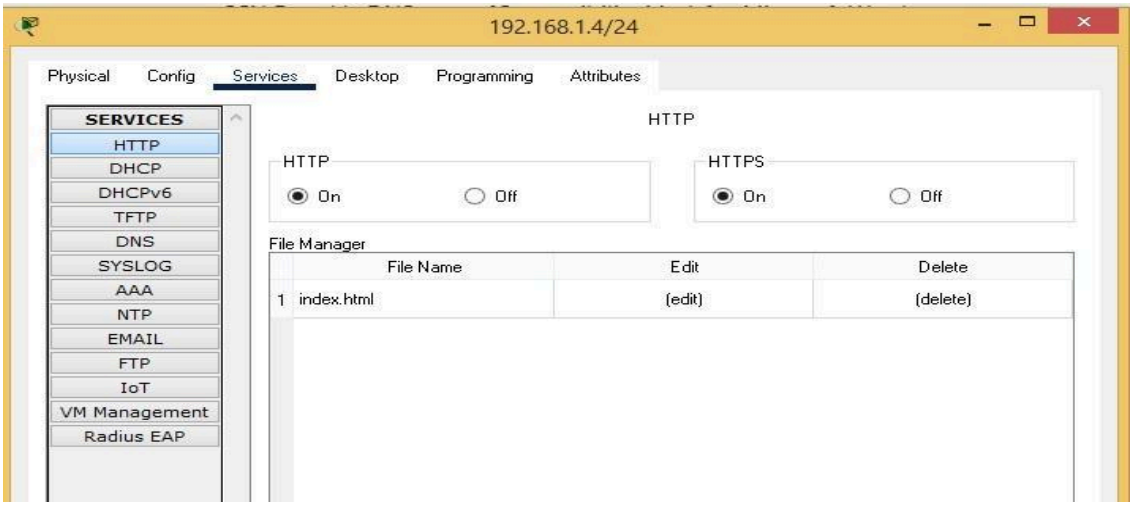
4. Go to each PC and add the IP address of the DNS server

E.g:

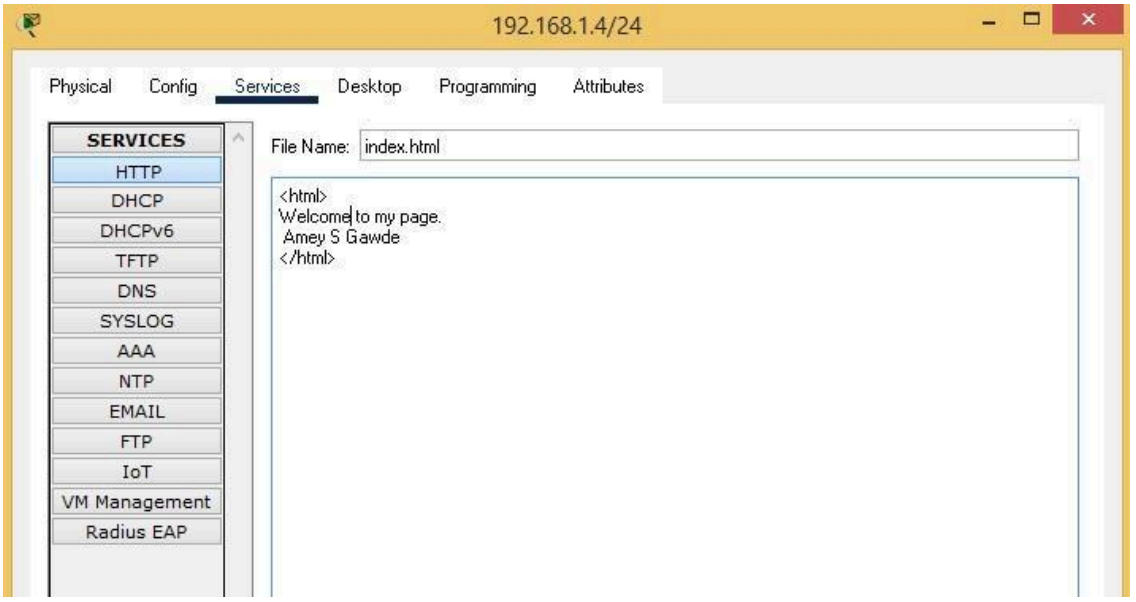


5. Ping from each PC to the DNS server.

6. Go to server and then to HTTP. Delete all the files just keep index.html as shown below



7. Edit the content of the index.html file with any other content. For example

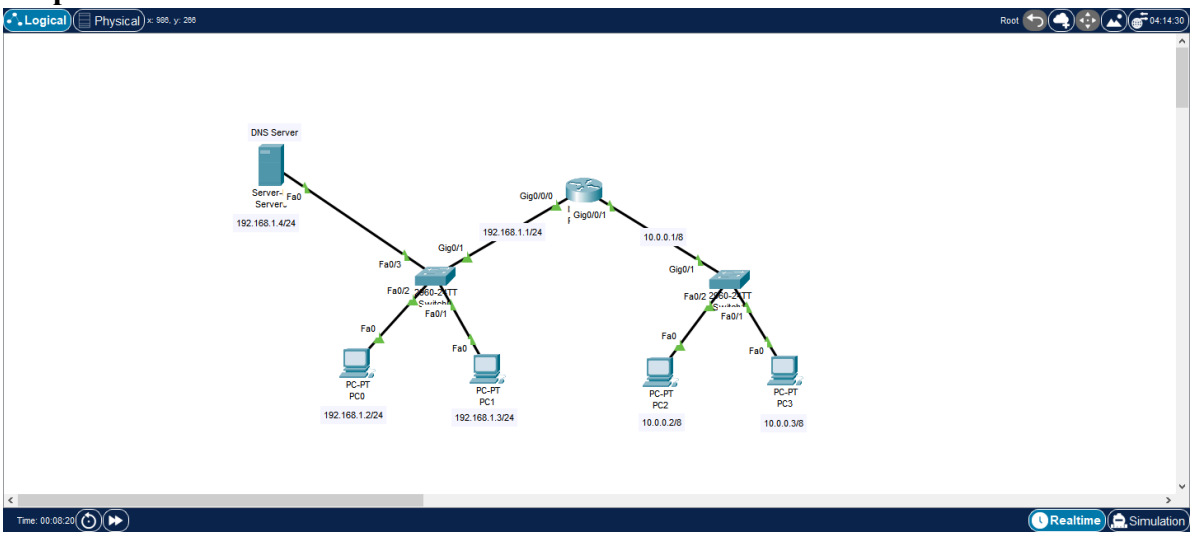


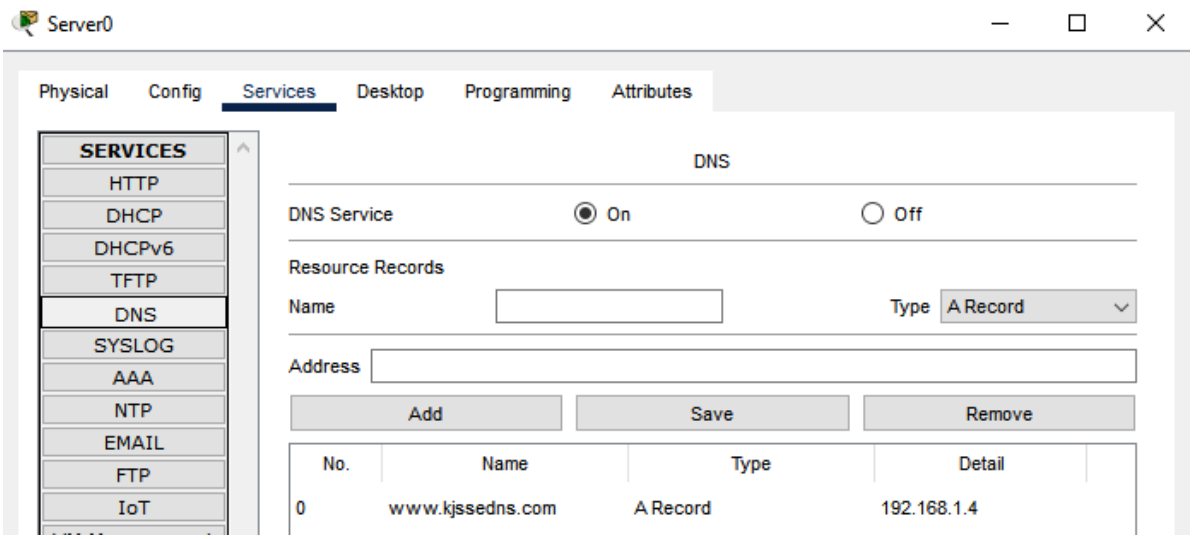
8. **PC-> Web Browser -> www.mywebpage.com and click on GO.**
It should open a page as shown below as shown below

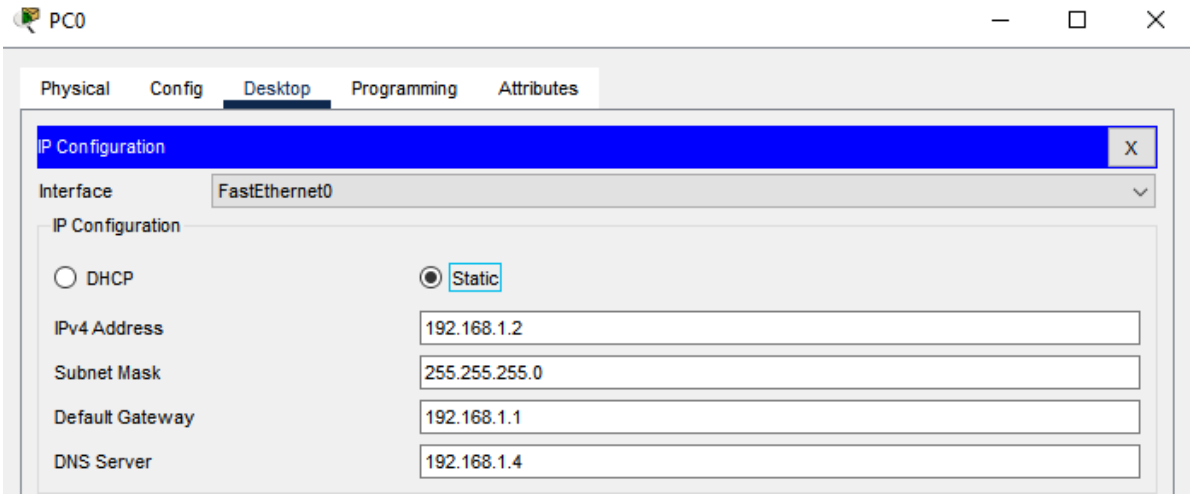


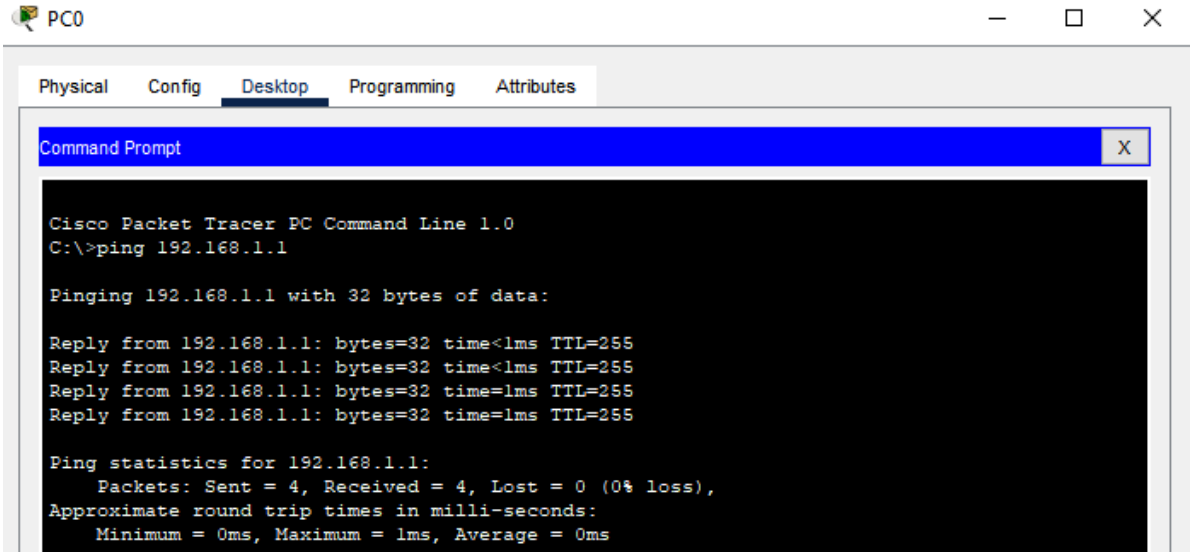
9. Go to all other PCs and check step 6 on all these PC

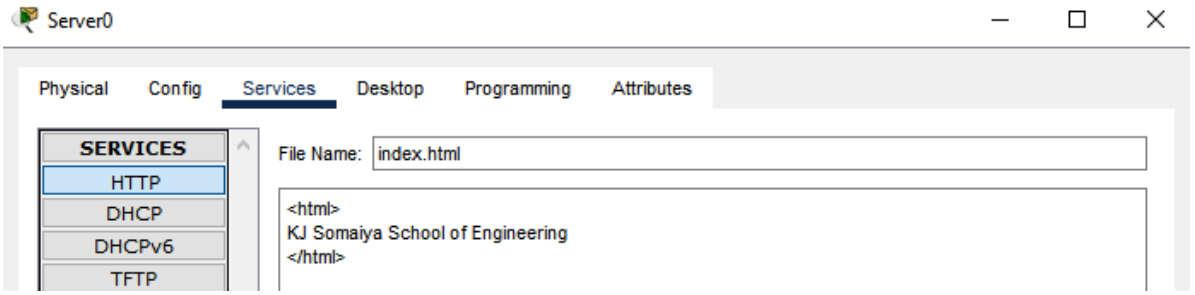
Output:

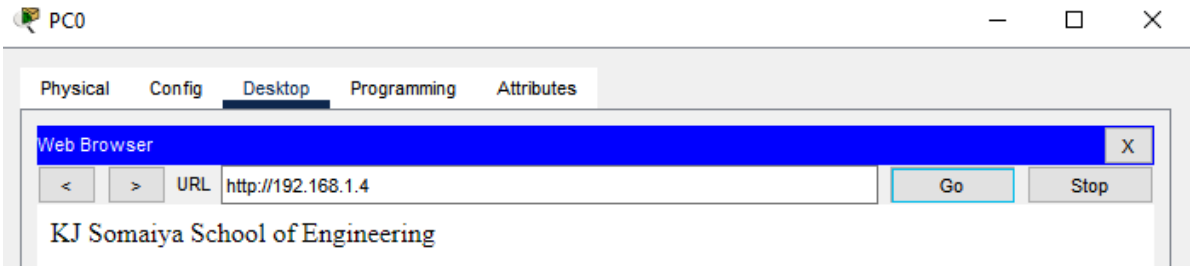


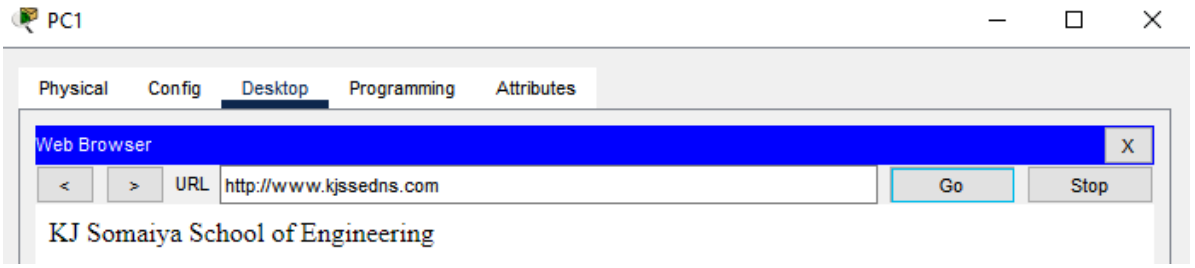



PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
192.168.1.4


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
File Name:
index.html
<html>
KJ Somaiya School of Engineering
</html>


PC0
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://192.168.1.4
Go
Stop
KJ Somaiya School of Engineering



CONCLUSION

In this experiment, I successfully configured the DNS server using Cisco Packet Tracer. The DNS service was enabled and an A record was created to map the domain name `www.mywebpage.com` to the IP address `192.168.1.4`. The DNS server address was configured on the PCs and connectivity was verified by pinging the DNS server. The webpage hosted on the HTTP server was successfully accessed using the domain name through the web browser. Thus, the experiment demonstrated the working of DNS and its role in translating domain names into IP addresses in a network.

Signature of faculty in-charge